LESSON 3-4 · DIVIDING POLYNOMIALS

Choose Your Route: Practice

Student ..... **Class** ..... **Date** ..... / ..... / 2026 **Teacher** Mr Malek Thiab

Three routes, one destination. Start anywhere on pages 1 and 2, move between routes whenever you like, and finish with page 3. A route names the task, not you.

PRACTICE · SECURE THE METHOD

Cards 1 to 3: divide with synthetic division and write the quotient and remainder. Cards 4 and 5: find the remainder with the Remainder Theorem. Card 6: decide if the divisor is a factor. Use the frame on the right.

1 Synthetic division

$(x^3 + 4x^2 - 5x - 14) \div (x - 2)$

2 Synthetic division

$(x^3 - 6x^2 + 11x - 2) \div (x - 3)$

3 Synthetic division

$(2x^3 + 3x^2 - 8x + 3) \div (x + 3)$

4 Remainder p(a)

$p(x) = x^3 - 5x + 7, \quad \div (x - 2)$

5 Remainder p(a)

$p(x) = 2x^3 + x^2 - 4, \quad \div (x + 1)$

6 Test, then factor

Is $x - 4$ a factor of $x^3 - 3x^2 - 10x + 24$?

Working space. For $x - a$ write a in the box. For $x + 3$ write -3. A missing power needs a zero coefficient.

Synthetic division frame. Write the box number, the coefficients, then bring down, multiply and add.

box | coeffs					
multiply					
add					

Two frames: use the second for a different card.

box | coeffs					
multiply					
add					

FIND THE ERROR

A student divided $x^3 + 2x^2 - 5x + 1$ by $x + 3$ and wrote the two lines above. Say what went wrong and write the correct quotient and remainder.

Box 3 : $\quad 1,\ 5,\ 10,\ 31 \ \Rightarrow$ remainder 31

so $(x^3 + 2x^2 - 5x + 1) \div (x + 3)$ has remainder 31

Done when... the box number has the opposite sign to the constant in the divisor, the last number is read as the remainder, and a zero remainder is linked to a factor.

LESSON 3-4 · DIVIDING POLYNOMIALS

Choose Your Route: Apply and Investigate

Algebra II · Choose Your Route

Student ..... **Class** ..... **Date** ..... / ..... / 2026 **Teacher** Mr Malek Thiab

APPLY · USE IT IN CONTEXT

1. A model box has volume $V(x) = x^3 + 6x^2 + 11x + 6$ cm ^{3} and height $(x + 1)$ cm. Find the base area $\dfrac{V(x)}{x+1}$.

2. For $p(x) = x^3 - 7x + k$, find k so that $x - 3$ is a factor.

3. Rewrite $\dfrac{x^2 + 7x + 9}{x + 3}$ in the form $q(x) + \dfrac{r}{x+3}$.

4. When $p(x) = x^3 + ax^2 - x + 2$ is divided by $x - 2$ the remainder is 12. Find a.

Done when: the divisor is written as $x - a$ first, the Remainder Theorem is used when only the remainder is needed, and each answer is checked.

INVESTIGATE · FIND OUT WHY

1. Write $p(x) = (x - a)\,q(x) + r$ and put $x = a$. Show that $r = p(a)$, then explain why this makes the Factor Theorem true.

2. Factor $x^3 - 2x^2 - 5x + 6$ fully. Use $p(1) = 0$ to find the first factor, then divide.

3. Without dividing, find the remainder when $x^{100} - 1$ is divided by $x - 1$, and when $x^{100} + 1$ is divided by $x + 1$.

Done when: your argument starts from $p(x) = (x - a)q(x) + r$ or from $p(a)$, and every remainder is explained.

SPOTLIGHT · AN IDEA THAT IS MY OWN

A route nobody else used, a question nobody else asked, or a surprise you found:

LESSON 3-4 · DIVIDING POLYNOMIALS

Which Route? Judge the Claims

Algebra II · Choose Your Route

Student ..... **Class** ..... **Date** ..... / ..... / 2026 **Teacher** Mr Malek Thiab

COMPARE THE ROUTES

Divide $x^3 - 5x^2 + 2x + 8$ by $x - 4$ three ways. Then compare: which was quickest, and which would you trust to show a factor?

ROUTE	my working	steps	where a slip could happen
Long division			
Synthetic division			
Remainder Theorem, then divide			

My choice and the evidence for it:

JUDGE THE CLAIMS

Decide true, false or sometimes true. One counterexample is enough to show a claim is false; a few examples are not enough to show it is true.

A	**Synthetic division works for any linear divisor, such as $2x + 3$.**	☐ true ☐ false ☐ sometimes	*My evidence:* *Assumption I checked:* Did I check the divisor is written $x - a$ with leading coefficient 1 ?
B	**If $p(a) = 0$ then $x - a$ is a factor of $p(x)$.**	☐ true ☐ false ☐ sometimes	*My evidence:* *Assumption I checked:* Did I test it on a polynomial where $p(a) \neq 0$ as well?
C	**A cubic divided by $x - a$ always gives a quadratic quotient.**	☐ true ☐ false ☐ sometimes	*My evidence:* *Assumption I checked:* Did I think about the degree of the quotient and the remainder?

WHAT IF...? PREDICT FIRST, THEN TEST

1. What if a power is missing, as in $x^3 - 7x + 6$? Predict the coefficient row you need, then divide by $x - 1$.

2. What if the divisor is $x + 1$? Predict the number in the box, then test it on $x^3 + 3x^2 - 4$.

3. Write a What if...? question of your own about dividing polynomials, predict the answer, and test it.

Done when... you have compared three routes with evidence, judged each claim with an example or counterexample, and tested at least one What if...? idea.